Milestone 5 — Physics-Informed AI Model Development (Step 6)

Deliverable of this step: *"Trained physics-informed AI models"* **Script:** `milestone5/run_milestone5.m` (log: `results/data/M5_summary.txt`; about 20 min in Octave, faster in MATLAB) **Models:** `results/models/pinet_fbg.mat`, `pinet_dense.mat`, `surrogate_hybrid.mat`, `M5_results.mat` **New code:** `src/pinn/` (PI-Net, physics decoder, evidence map, beam prior, hybrid surrogate), `src/ml/` (MLP, Adam, gradient check), `validation/val12_ai_components.m`

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1. Purpose — and what "physics-informed" means here

Step 6 of the workflow asks for ML/DL models whose loss contains the governing equations and boundary conditions, so that predictions stay physically consistent. Milestone 5 builds **two** physics-informed AI models. Both are used later by the digital twin (Milestone 6).

Model	Question it answers	Data
PI-Net (damage diagnosis)	*Is the wing damaged? What type, which face, where, how large, how much strength is left?*	SHM tier: 8 400 noisy sensor records (Milestone 4)
Hybrid surrogate (design / health-state model)	*What are f_1B, f_1T, tip deflection and strength of a wing with these design parameters?* in microseconds instead of 9 s of FE	Design tier: 400 FE wings (Milestone 3)

Physics enters these models in **five** distinct ways. Each one is tested separately in the ablation studies:

#	Mechanism	Where	Physics used
P1	**Healthy-referenced inputs**	PI-Net input layer	Damage = deviation from the healthy population; EOV-correlated directions are whitened (Mahalanobis)
P2	**Physics-consistency loss** (discrete PINN residual)	PI-Net loss	Discretised governing equations K u = F and K φ = λ M φ, through FE sensitivities
P3	**Physics evidence prior**	PI-Net map head	Modal-strain-energy and strain-residual damage indices (Milestone 4)
P4	**Reduced-order mechanics + learned model-form error**	Hybrid surrogate	Ritz beam on elastic root, CLT, Dunkerley, Tsai–Wu

#	Mechanism	Where	Physics used
P5	**Dimensionless CLT groups** as inputs	Hybrid surrogate	Bending-twist coupling D_{16}, D_{26}, orthotropy, core shear flexibility

Why not a "classic" collocation PINN? A classic PINN (Raissi et al. 2019) trains a network u(x; θ) so that the PDE residual vanishes at collocation points. It is a *PDE solver*. We already have an exact, validated FSDT solver (Milestone 2: 56/56 checks). The open problem is the **inverse** problem: from sparse, noisy sensors, infer the damage field. For the inverse problem, the most effective way to embed the governing equations is the **discrete** form, K(D) u = F. Its exact first-order sensitivity, computed by adjoint and eigenvalue-perturbation analysis, becomes a differentiable "physics decoder" inside the loss (P2). This keeps the full fidelity of the FE model, including the elastic root BC, the sandwich shear correction and the sensor positions. A collocation PINN would have to *re-learn* all of that.

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2. Theory

2.1 Feed-forward networks (ANN / MLP)

A multilayer perceptron maps inputs x through layers

$$\mathbf{a}^{(0)} = \mathbf{x}, \qquad \mathbf{a}^{(l)} = \tanh\left(\mathbf{a}^{(l-1)}\mathbf{W}^{(l)} + \mathbf{b}^{(l)}\right), \quad l = 1..L.$$

Output heads use the link function that matches each physical quantity:

Quantity	Link	Range
Class probabilities	softmax $p_c = e^{z_c} / \sum_k e^{z_k}$	simplex
Damaged face	sigmoid $\sigma(z) = 1/(1+e^{-z})$	(0, 1)
Damage map	softmax over the 400 elements	sums to 1
Damage extent	softplus $s = \ln(1+e^{z})$	≥ 0
Residual strength	linear	$\mathbb{R}$

- **Initialisation (Glorot & Bengio 2010):** $W \sim U(-\sqrt{6/(n_{in}+n_{out})}, +\sqrt{6/(n_{in}+n_{out})})$. This keeps the activation variance constant through tanh layers.
- **Optimiser — Adam (Kingma & Ba 2015):**

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \quad \theta_t = \theta_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

with bias corrections $\hat{m} = m/(1-\beta_1^t)$ and $\hat{v} = v/(1-\beta_2^t)$. The learning rate follows a cosine schedule from η_0 to $\eta_0/20$. - **Regularisation:** - L2 weight decay; - early stopping on the **validation** objective, with the best parameters kept; - **Gaussian input-noise augmentation** (σ = 0.6 in whitened units). The inputs *are* noisy measurements. Training with input noise is equivalent to Tikhonov regularisation of the input–output Jacobian (Bishop 1995), so the network cannot rely on exact feature values.

2.2 Back-propagation (hand-derived, no toolbox)

All gradients are derived and coded by hand. The key identities are:

Block	Gradient
softmax + cross-entropy	$\partial L/\partial z = p - y$
softmax (general upstream g)	$\partial L/\partial z = q \odot (g - q^T g)$
sigmoid + BCE	$\partial L/\partial z = p - t$
softplus	$ds/dz = \sigma(z)$
tanh layer	$\delta^{(l)} = (\delta^{(l+1)} W^{(l+1)T}) \odot (1 - a^{(l)2})$, $\partial L/\partial W^{(l)} = a^{(l-1)T} \delta^{(l)}$

They are verified against central finite differences:

$$\varepsilon_{rel} = \max_i \frac{|g_i - \tilde{g}_i|}{|g_i| + |\tilde{g}_i|}, \qquad \tilde{g}_i = \frac{L(\theta + he_i) - L(\theta - he_i)}{2h}, \; h = 10^{-6}.$$

Correct code gives about 10^{-7}. A single wrong term gives about 10^{-1}. This is validation check V12 in `run_all_validation`.

2.3 Healthy-referenced inputs (P1)

Damage is a **deviation from the healthy population**. That population is strongly correlated by environmental and operational variability (EOV): temperature, root fixity, mass and material scatter all move many features together (Milestone 3: EOV shifts frequencies 10–100× more than damage does). The 117 Milestone 4 features are therefore transformed with statistics of the **healthy training wings only**:

$$\mathbf{z} = \mathbf{S}_h^{-1/2}(\mathbf{x} - \boldsymbol{\mu}_h), \qquad \mathbf{S}_h \leftarrow 0.95\,\mathbf{S}_h + 0.05\,\mathrm{diag}(\mathbf{S}_h)$$

(symmetric inverse square root from the eigen-decomposition). Then $\|\mathbf{z}\|^2$ *is* the Mahalanobis novelty index of Milestone 4.

Six family indices $\ln(1 + D_g^2/n_g)$ are appended, one each for frequency, ratio, MAC, ULS, strain and MSE index. Temperature is excluded because it is EOV, not damage. This gives 123 inputs.

The network therefore starts from the optimal *unsupervised* detector. It only has to learn what statistics cannot: which **pattern** of deviations belongs to which damage type, where and how large.

2.4 PI-Net — architecture and multi-task loss

```
 123 inputs → tanh(64) → tanh(64) → h ─┬→ softmax(7)      p_c   class: healthy,
debond, matrix, fibre, crush, delam, root
                                      ├→ sigmoid(1)      p_top damaged face
                                      ├→ softmax(400)    q_e   damage map   (+ a·E
physics evidence prior, P3)
                                      ├→ softplus(1)     s     extent  Σ D_e / 5
                                      └→ linear(1)       rf    residual strength
RF_min (standardised)
 decoder (P2):  d̂ = 5 s q  ⟶  x̂ = Σ_t π_t J_t d̂ + π_root J_root 5s  ⟶  compare with
the measured Δx
```

Supervised losses (only on labelled wings):

$$L_{cls} = -\tfrac{1}{N}\sum \ln p_y, \quad L_{face} = \mathrm{BCE}(p_{top}, t), \quad L_{map} = -\tfrac{1}{N}\sum_e q_e^* \ln q_e, \quad L_{loc} = \tfrac{1}{N\,\ell^2}\sum \Big\| \sum_e q_e \mathbf{x}_e - \mathbf{x}^* \Big\|^2,$$
$$L_{ext} = \tfrac{1}{N}\sum (s - s^*)^2, \quad L_{rf} = \tfrac{1}{N}\sum (rf - rf^*)^2$$

- q* = D/ΣD is the true (normalised) damage map, and ℓ = 0.1 m.
- Face labels exist only for skin and delamination damage; map and location labels only for localisable classes.
- The total loss is $L = \sum_k w_k L_k + \lambda_{phys} L_{phys} + \lambda_2 \| W \|^2$, with w = (1, 0.3, 0.3, 1, 0.3, 0.3), λ_phys = 0.3 and $\lambda_2 = 10^{-2}$.

2.5 The physics decoder and the physics-consistency loss (P2)

For every damage type t and element e, the first-order change of the measurable features per unit damage severity is computed **from the discretised governing equations** of the healthy reference wing:

Natural frequencies — eigenvalue perturbation (K φ = λ M φ, φᵀMφ = 1; damage changes only K):

$$\frac{\partial \lambda_k}{\partial D_e} = \boldsymbol{\phi}_k^T \frac{\partial \mathbf{K}_e}{\partial D} \boldsymbol{\phi}_k, \qquad \frac{\partial f_k}{f_k} = \frac{1}{2} \frac{\partial \lambda_k}{\lambda_k}, \qquad \Delta\left(\frac{f_k}{f_1}\right) = \frac{f_k}{f_1}\left(\frac{\partial f_k}{f_k} - \frac{\partial f_1}{f_1}\right)$$

Static tip-test strains — direct (adjoint) differentiation of K u = F:

$$\frac{\partial \mathbf{u}}{\partial D_e} = -\mathbf{K}^{-1} \frac{\partial \mathbf{K}_e}{\partial D} \mathbf{u}, \qquad \Delta\varepsilon_s = \left(\mathbf{B}_m \mp \tfrac{h}{2} \mathbf{B}_b\right)_s \Delta\mathbf{u}$$

at all 60 FBGs (bottom −, top +). Then the **same** global-scale removal as the Milestone 4 feature:

$$\Delta\mathbf{x} = \frac{1}{\max |\varepsilon_0|} \left(\mathbf{I} - \frac{\varepsilon_0 \varepsilon_0^T}{\varepsilon_0^T \varepsilon_0}\right) \Delta\varepsilon.$$

∂K_e/∂D is a forward difference of the *element* stiffness, using the damage module's own degradation rules. The decoder therefore knows exactly what each damage type does:

- a debond softens core shear;
- matrix cracks reduce E_2 and G_{12} of one face;
- fibre breakage reduces E_1 of one face;
- crushing reduces core moduli;
- delamination reduces interlaminar shear coupling;
- root loosening reduces the root springs, including the elastic-root **boundary condition**.

Stacking all elements gives J_t (64 × 400) for 8 types, plus J_root.

Physics-consistency loss. The network's diagnosis is turned into a predicted damage field $\hat{d}_e = 5 s q_e$ and pushed through the decoder, with the type probabilities π_t built from p_c and p_top:

$$\hat{\mathbf{x}} = \sum_t \pi_t \mathbf{J}_t \hat{\mathbf{d}} + \pi_{root} \mathbf{J}_{root}\, 5s, \qquad L_{phys} = \frac{1}{N n_f} \sum_n \left\| \frac{\hat{\mathbf{x}}_n - \Delta\mathbf{x}_n}{\boldsymbol{\sigma}} \right\|^2.$$

In words: **the damage the network claims to see must, through the governing equations, reproduce what the sensors measured.** This is the discrete analogue of a PINN residual. It has two important properties:

1. **It needs no labels.** L_phys is evaluated on *all* wings, so unlabelled wings (the normal case in service) still train the network (semi-supervised learning).
2. **It couples the heads physically.** Class, face, map and extent must be mutually consistent, because only a consistent combination reproduces the measurement.

2.6 Physics evidence-map prior (P3)

The map head computes

$$q_e = \mathrm{softmax}_e(\mathbf{W}\mathbf{h} + \mathbf{b} + a E_e),$$

- E is the Milestone 4 physics damage-evidence map, healthy-z-scored:
  - **FBG variant:** the MSE index plus the |strain residual| of both skins, at 30 grating positions, spread to elements with a Gaussian kernel (ℓ = 60 mm);
  - **dense variant:** the full-field z-map, e.g. from DIC.
- a is a learnable gain.

In Bayesian terms, $\ln q_e = \ln(\text{physics evidence}) + \text{learned correction} + \text{const}$. The network *refines* a physics-based localisation instead of learning the geometry of the wing from 1 960 examples (residual or prior learning).

2.7 Hybrid physics-AI surrogate (P4, P5)

Reduced-order mechanics (`beam_prior.m`, about 10 ms per design). The plate wing is condensed into a tapered cantilever on an elastic root:

- **Section properties (CLT):**
  - $EI(x) = c(x)/d_{11}$, with d = A^{-1} of the full ABD matrix (wide-beam, free anticlastic curvature);
  - $kGA = c\,A_{s,11}$ (energy-equivalent k of Milestone 2);
  - $GJ = 4D_{66}c$;
  - $m = I_0 c$;
  - root springs $K_\theta = k^* D_{11} c_r / L$ (identical to the FE model).
- **f_1B — Rayleigh–Ritz.** Use $w = \sum_{j=1}^{6} a_j (x/L)^j$ and solve **Ka** = ω^2 **Ma** with

$$K_{ij} = \int_0^L EI\,\phi_i''\phi_j''\,dx + K_\theta \phi_i'(0)\phi_j'(0), \qquad M_{ij} = \int_0^L m\,\phi_i\phi_j\,dx.$$

The j = 1 term is the rigid rotation permitted by the root spring. Core shear is added with **Dunkerley's rule**, $f^{-2} = f_b^{-2} + f_s^{-2}$, where $f_s = \frac{1}{4L}\sqrt{kGA/m}$. - **$f_1$T:** uniform torsion $\frac{1}{4L}\sqrt{GJ/I_p}$, in series with the root torsion spring (Dunkerley). - **Tip deflection:** Ritz statics under the net 1 g load of the FE model (Schrenk lift − weight), plus the shear deflection $\int V/kGA\,dx$. - **RF (Tsai–Wu):** ultimate bending moment at x_off; ply strains $\varepsilon = \varepsilon^0 + z\kappa$ from the ABD inverse; then Tsai–Wu strength ratio of every face ply (first-ply failure).

The model **deliberately ignores** sweep (bending-torsion coupling), bend-twist coupling D_{16}/D_{26}, 2-D load diffusion at the root, chordwise bending and the true pressure distribution. That missing physics is the *model-form error*.

Discrepancy learning (Kennedy & O'Hagan 2001):

$$\ln y_{FE}(\mathbf{x}) = \ln y_{phys}(\mathbf{x}) + \delta_{NN}(\mathbf{x},\ \boldsymbol{\pi}_{CLT}(\mathbf{x}),\ \ln y_{phys}(\mathbf{x}))$$

The network learns only the *correction*. With no data the model falls back to physics. The inputs are:

- the 17 encoded design variables. Ply angles enter as (cos 2θ, sin 2θ), because a 0° and a 180° ply are the same ply.
- **six dimensionless CLT groups π_CLT (P5):** ln D_{22}/D_{11}, ln D_{66}/D_{11}, D_{12}/D_{11}, $D_{16}/\sqrt{(D_{11}D_{66})}$, $D_{26}/\sqrt{(D_{22}D_{66})}$ and ln $D_{11}/(kGA\,L^2)$. These are the similarity parameters of anisotropic plate mechanics. Variables with physical meaning generalise far better than raw angles.

Three alternatives are compared, all with identical networks, training and validation-based L2 selection:

- the pure network;
- the network plus CLT groups;
- the hybrid.

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3. Verification of the building blocks

3.1 Gradients and reduced-order model (V12, part of `run_all_validation`)

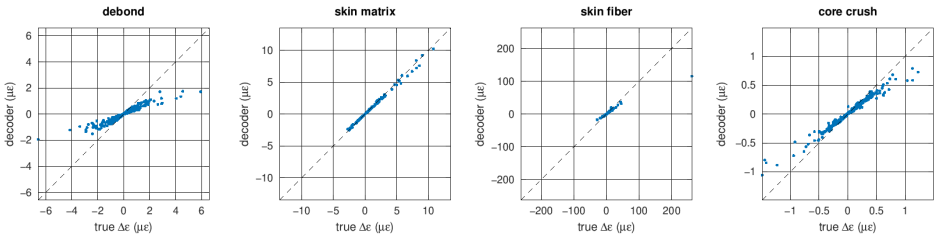
Check	Result	Tolerance	Status
PI-Net gradient, all 7 loss terms incl. physics decoder and evidence gain	max rel. error 2.2×10^{-7}	10^{-5}	PASS
Surrogate MLP gradient	3.6×10^{-7}	10^{-5}	PASS
Beam/CLT prior f_1B vs FE (baseline wing)	30.61 vs 30.88 Hz (0.9 %)	5 %	PASS
Beam/CLT prior 1 g tip deflection vs FE	1.174 vs 1.157 mm (1.4 %)	5 %	PASS

The validation suite now has **60 required checks, and all pass**.

3.2 Physics decoder vs exact nonlinear twins

The decoder is a *linearisation*, so it is checked against the noise-free "twin" pairs of Milestone 3 (the same wing simulated with and without damage, 60 per class). The predicted FBG strain change $\mathbf{J}_t\mathbf{D}$ is compared with the exact change:

Damage	cosine(predicted, true) median (p10)	amplitude ratio	true signal $\|\Delta\varepsilon\|$
Core debond	0.999 (0.993)	0.75	1.18 με
Skin matrix cracks	0.999 (0.963)	0.96	0.86 με
Skin fibre breakage	0.999 (0.985)	0.85	2.57 με
Core crush	0.996 (0.982)	0.86	0.43 με
Delamination	—	—	< 0.001 με, invisible to static strain



Decoder validation

Interpretation:

- The decoder reproduces the **shape** of the strain change almost exactly (cosine ≈ 0.999).
- It under-predicts the **amplitude** of large damage by 4–25 %. Stiffness loss is nonlinear in D: a strongly softened element sheds load to its neighbours.
- The most important number is the last column. The whole damage signature in the 60-FBG tip test is **0.4–2.6 με**. The FBG noise is 1 με per grating, plus 0.5 % gain and 1 % load error. The physics is right, but the information available to *any* model, physics-informed or not, is close to the noise floor. Every result below must be read against this.

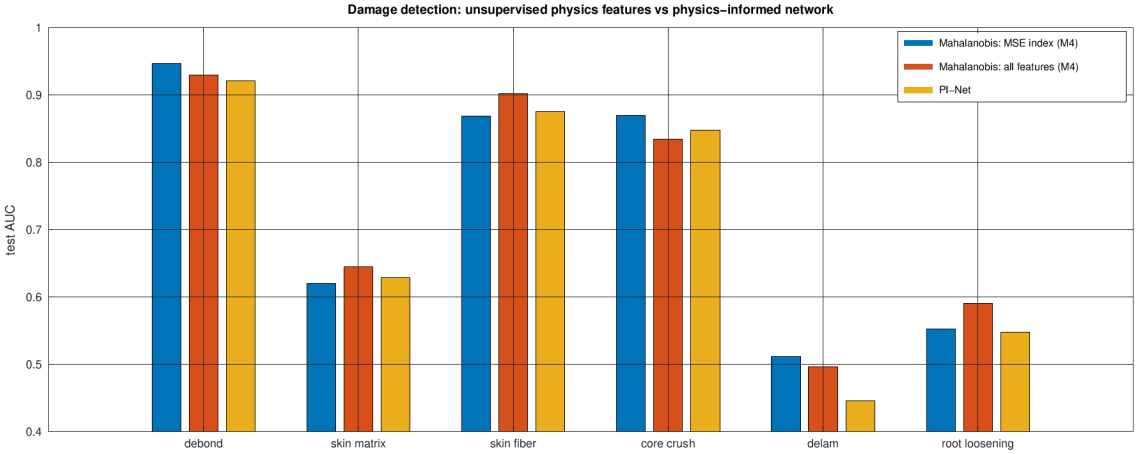
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4. PI-Net results (untouched test set: 420 wings, 1 260 records)

Model: 123 → 64 → 64 → heads, 38 747 parameters. It was trained in 36 s (63 epochs, best validation at epoch 23), with λ_phys = 0.3 and the FBG evidence prior. All choices (inputs, width, L2, noise level, λ, prior) were made on the **validation** set; the test set was used once.

4.1 Detection (healthy vs each damage type)

Method	debond	skin matrix	skin fibre	core crush	delam	root loosening
AUC — Mahalanobis, MSE index (M4)	**0.95**	0.62	0.87	**0.87**	0.51	0.55
AUC — Mahalanobis, all features (M4)	0.93	**0.65**	**0.90**	0.83	0.50	**0.59**
AUC — **PI-Net** (1 − p_healthy)	0.92	0.63	0.88	0.85	0.45	0.55
Detection rate @ 1 % FA — MSE index (M4)	81 %	18 %	50 %	62 %	3 %	16 %
Detection rate @ 1 % FA — **PI-Net**	**86 %**	**27 %**	**73 %**	**63 %**	3 %	**21 %**



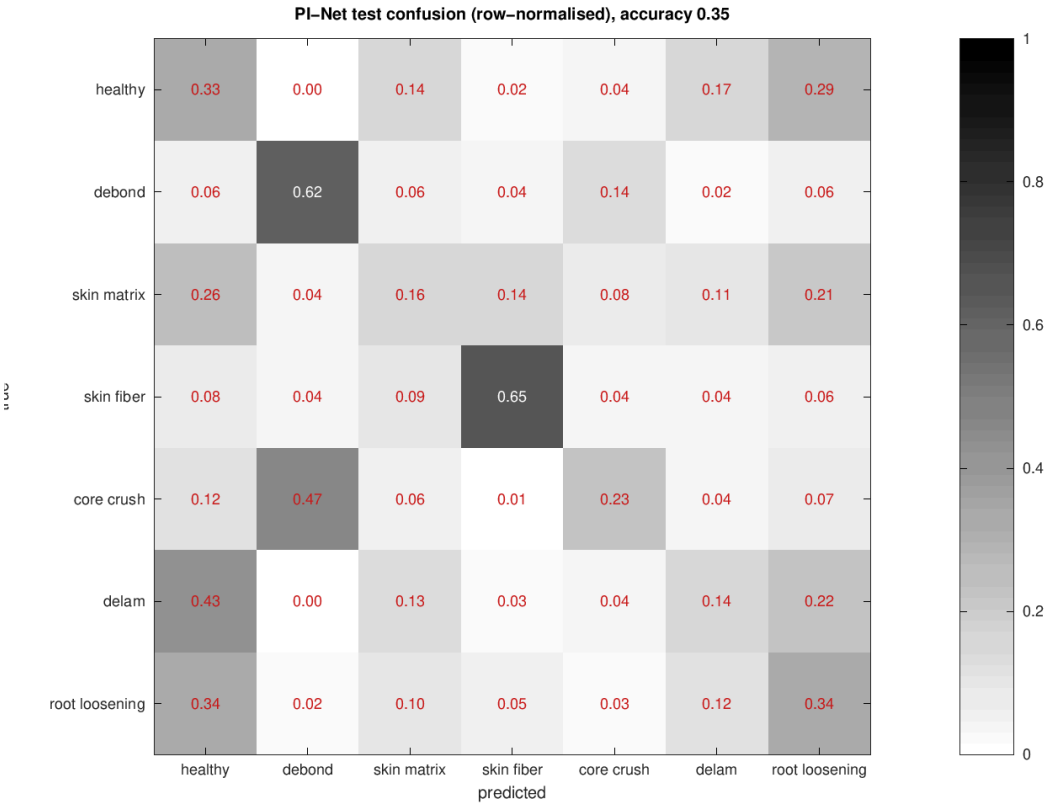
Detection AUC

Interpretation:

- **Overall ranking (AUC):** PI-Net equals the best unsupervised detector, within ±0.03.
- **Detection at an operational false-alarm level:** PI-Net detects clearly more. Fibre-break detection rises from 50 % to **73 %**, matrix cracks from 18 % to 27 %, and debonds from 81 % to 86 %. The network combines the families selectively, while the “all features” Mahalanobis distance dilutes them (Milestone 4, finding 4).
- **Measured false-alarm rate:** 3.3 % on the test set, against the 1 % target set on validation. With 180 healthy validation records, the 99th percentile is estimated from only about 2 exceedances. Milestone 6 will calibrate the threshold on a larger healthy baseline using extreme-value statistics.
- **Delamination** stays at chance level, consistent with the < 0.001 με signal above.

4.2 Damage-type diagnosis (7 classes)

Accuracy 0.355, macro-F1 0.349 (chance 0.143). Per-class F1: fibre 0.67, debond 0.57, root loosening 0.30, core crush 0.29, healthy 0.26, delam 0.18, matrix 0.18.



Confusion

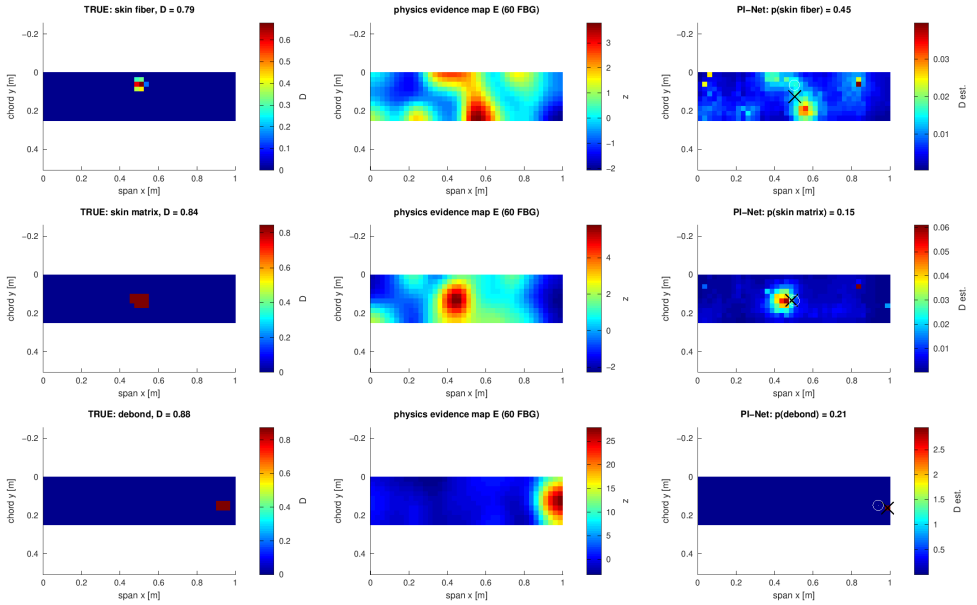
The confusion matrix has a clear **physical structure**:

- **Core crush → debond (44 %).** Both degrade the core and produce nearly the same modal-strain-energy pattern, which is a genuine physical ambiguity.
- **Healthy ↔ delamination ↔ root loosening ↔ matrix** form one confusion block. These are the four "invisible" states: their signals are below the EOV scatter, so the network spreads probability among them.
- **Fibre breakage and debonding** are the two classes with strong, distinctive signatures, and they are recognised (67 % and 59 % recall).

This is the honest information limit of a 60-FBG + 12-accelerometer network under realistic EOV. It is *not* a training failure: the same limits appeared in the purely physical analysis of Milestones 3–4.

4.3 Localisation (significant damage: D > 0.5, radius > 30 mm)

Class	M4 rule, 60 FBG	PI-Net, 60 FBG	M4 rule, dense	PI-Net, dense
Debond	92 mm	84 mm	59 mm	64 mm
Skin matrix	**67 mm**	90 mm	**21 mm**	58 mm
Skin fibre	**24 mm**	37 mm	6 mm	8 mm (98 % < 50 mm)
Core crush	161 mm	**99 mm**	177 mm	**94 mm**



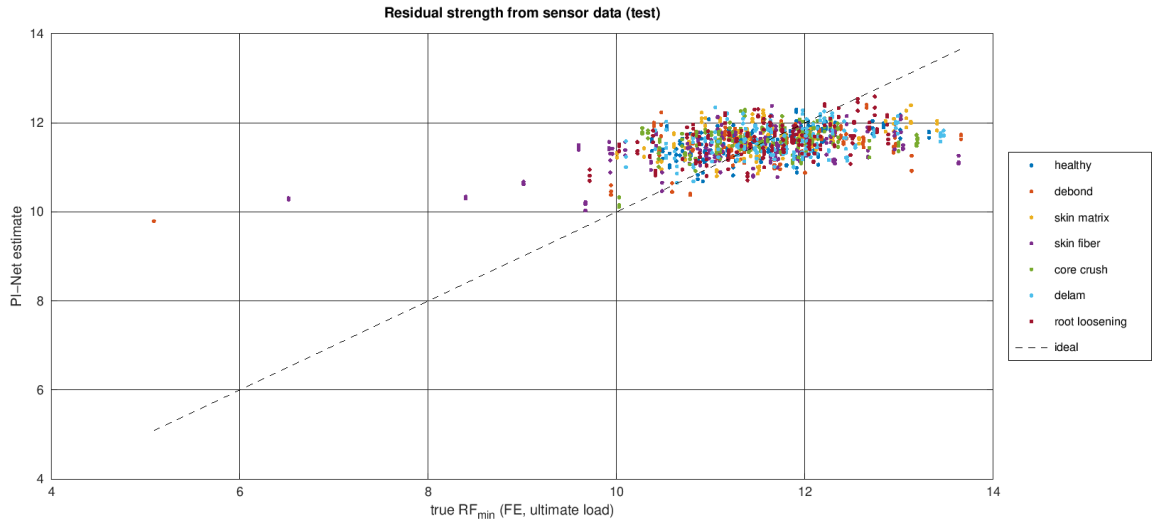
Damage maps

Interpretation:

- **Core crush improves the most** (161 → 99 mm with FBGs, 177 → 94 mm dense). The rule-based method fails because crush leaves almost no surface-strain trace. The network learns to combine the weak modal evidence with the strain map, so the multivariate model adds information here.
- **For skin damage the simple M4 rule is still sharper.** Its "top-k sensors" readout is a near-optimal estimator when the evidence is a sharp peak at the sensor spacing (100 × 75 mm). The network reports the centroid of a *probability map*, which is smoother.
- This is why both are kept. The digital twin (M6) uses the PI-Net map as a **posterior** and the rule as a point estimate.
- The damage-map panels show what the network outputs. For the fibre break (top row), the evidence map is diffuse. The network's map concentrates near the true site, but with only p = 0.47 confidence. That uncertainty is honest.

4.4 Quantification and residual strength

Quantity	Result
Damaged face (skin damage)	64 % correct
Damage extent ΣD	Spearman ρ = 0.26, $R^2 \approx 0$
Residual strength RF_min	RMSE 0.76 (R^2 0.22); clearly weakened wings: RMSE 1.87



RF parity

Honest conclusion: *direct* regression of extent and strength from noisy global features does **not** work well. This is not a modelling detail. Strength is governed by the **local** stress concentration at the damage, while the sensors see the **global** stiffness change, which is 1–3 με. Two wings with the same sensor signature can have very different RF.

This result decides the architecture of Milestone 6. The twin will **not** regress RF from features. It will:

1. use PI-Net to estimate type, location and the damage-map posterior;
2. insert that damage into the **FE model** (model updating);
3. compute RF with Tsai–Wu and Hashin, i.e. physics.

This is the standard "diagnosis → model update → prognosis" chain of a structural digital twin (Farrar & Worden 2013), and it is far more reliable for strength.

4.5 Physics consistency

Mean squared residual between the decoded and the measured feature changes on the test set, in units of the healthy σ:

Prediction	Residual
"No damage"	2.99
PI-Net with λ_phys = 0	1.81
PI-Net with λ_phys = 0.3	**1.58**
PI-Net with λ_phys = 1	1.57

With the physics loss, the network's diagnoses explain **47 %** of the measured feature change through the governing equations, against 39 % without it. Its outputs are *physically consistent* explanations of the data, not just labels.

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5. Ablation studies — what each physics ingredient contributes

5.1 Healthy-referenced inputs (P1) — the largest single gain

Development history, **validation** set, identical network (64-64, L2 10^{-2}, input noise 0.6):

Inputs	Accuracy	AUC debond / matrix / fibre / crush
Standardised features (z-score over all training data)	0.247	0.83 / 0.66 / 0.69 / 0.78
Healthy-whitened + family novelty indices	**0.350**	**0.93 / 0.74 / 0.88 / 0.82**

The first networks were **worse than the unsupervised Mahalanobis detector**. The reason: with about 2 000 training wings, a network cannot discover the 117 × 117 EOV covariance structure by itself. Giving it the healthy reference explicitly is a physics/statistics prior, and it raised accuracy by 40 % (relative).

5.2 Evidence-map prior (P3), test set

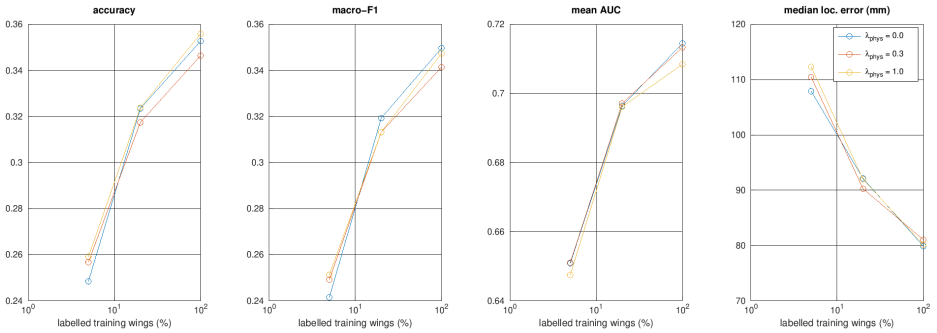
Variant	Accuracy	Median localisation error, debond / matrix / fibre / crush	Hit < 50 mm
No prior	0.347	73 / 129 / 102 / 89 mm	18 %
FBG evidence prior	0.355	84 / 90 / **37** / 99 mm	32 %
Dense evidence prior	0.359	64 / **58** / **8** / 94 mm	**51 %**

- The prior is what makes **skin-damage localisation** work: the fibre-break error drops from 102 to 37 mm.
- The learned gain a ≈ 0.5 shows that the network trusts the physics map but also corrects it.
- The prior does not change the classification.

5.3 Physics loss (P2) × labelled fraction — semi-supervised learning

Each cell is the mean of 2 seeds on the test set. Unlabelled training wings contribute only through L_phys.

Labelled wings	λ_phys	Accuracy	Macro-F1	Mean AUC	Physics residual
5 %	0	0.248	0.241	0.651	2.45
5 %	0.3	0.257	0.249	0.651	2.13
5 %	1.0	**0.259**	**0.251**	0.647	**2.00**
20 %	0	0.323	0.319	0.696	2.39
20 %	0.3	0.317	0.313	0.697	1.80
20 %	1.0	0.324	0.313	0.696	1.73
100 %	0	0.353	0.350	0.714	1.81
100 %	0.3	0.346	0.341	0.713	1.58
100 %	1.0	0.356	0.347	0.708	1.57



Label efficiency

Honest reading:

1. The physics loss **always** makes predictions more physically consistent: the residual falls by 13–28 %.
2. It gives a **small accuracy benefit only in the scarce-label regime** (5 % labels: +1 point accuracy and macro-F1). With 20–100 % labels, the effect (±0.01) is inside the seed-to-seed scatter.
3. The reason is §3.2. The physics loss can only transfer information that is *in* the measurement. With a damage signature of 0.4–2.6 με against about 1 με noise, unlabelled data contain little extra damage information. The literature reports large PINN gains in settings with rich, low-noise data (full-field, many sensors). Our study quantifies why such gains do not automatically carry over to a sparse, realistic SHM network.

This is a **negative result**, reported because it is true and useful. The physics loss is retained (λ = 0.3) for consistency and the low-label benefit, not because it raises accuracy.

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6. Hybrid physics-AI surrogate (design tier)

6.1 Reduced-order model alone

Over 400 random designs (ply angles ±90°, ply 0.10–0.20 mm, core 8–30 mm, foam grade 0.5–2, 3 face materials, taper 0.5–1, sweep 0–20°, root fixity k* 200–5 000):

Output	Median FE / model	corr(log)	RMS error
f_1B	1.004	0.981	7.9 %
f_1T	0.750	0.904	36.6 %
w_tip (1 g)	0.984	0.991	12.0 %
RF Tsai–Wu	0.964	0.983	16.6 %

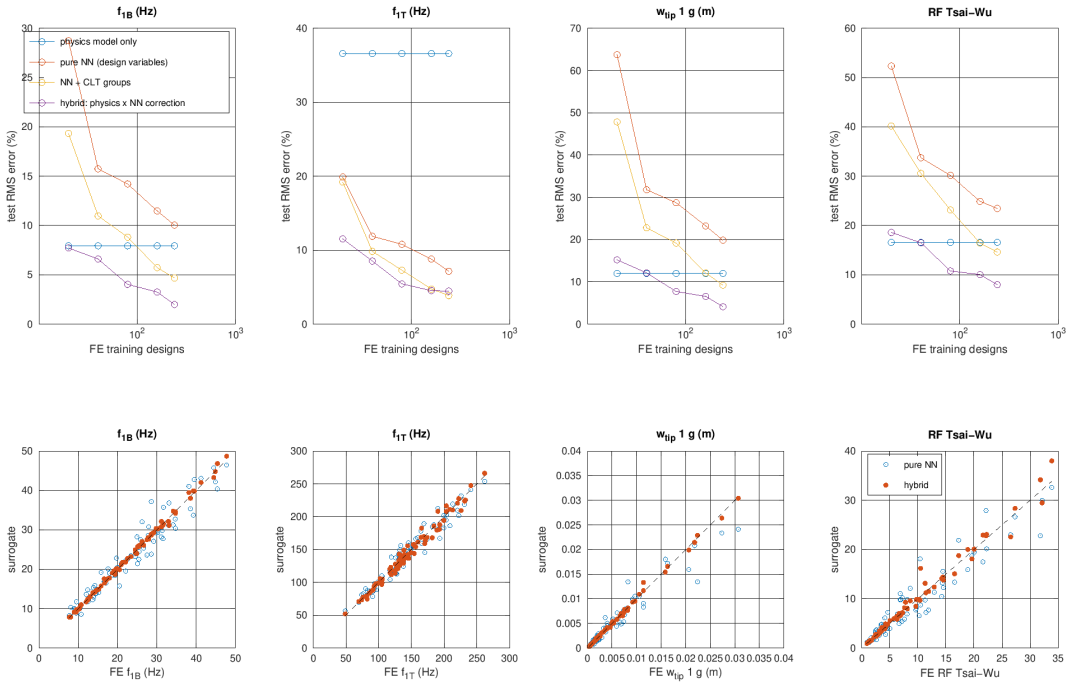
The 10 ms beam/CLT model already captures most of the physics. Torsion is the weakest output (−25 % bias): the Kirchhoff plate estimate GJ = $4D_{66}c$ ignores core shear in torsion.

6.2 Learning curves — test RMS error vs number of FE training designs

80 unseen test designs, 3 random splits:

n FE designs	Physics only	Pure NN	NN + CLT groups	Hybrid
20	7.9 / 36.6 / 12.0 / 16.6	28.7 / 19.9 / 63.7 / 52.3	19.3 / 19.2 / 47.8 / 40.2	**7.7 / 11.5 / 15.2 / 18.6**
40	〃	15.7 / 11.9 / 31.8 / 33.7	11.0 / 9.8 / 22.8 / 30.6	**6.6 / 8.5 / 12.1 / 16.5**
80	〃	14.2 / 10.8 / 28.7 / 30.2	8.8 / 7.3 / 19.2 / 23.1	**4.0 / 5.4 / 7.7 / 10.8**
160	〃	11.5 / 8.8 / 23.2 / 24.9	5.7 / 4.7 / 12.1 / 16.4	**3.2 / 4.5 / 6.6 / 10.0**
240	〃	10.0 / 7.1 / 19.8 / 23.5	4.7 / **3.9** / 9.2 / 14.6	**2.0** / 4.5 / **4.1** / **8.0**

(Values are % for f_1B / f_1T / w_tip / RF.)



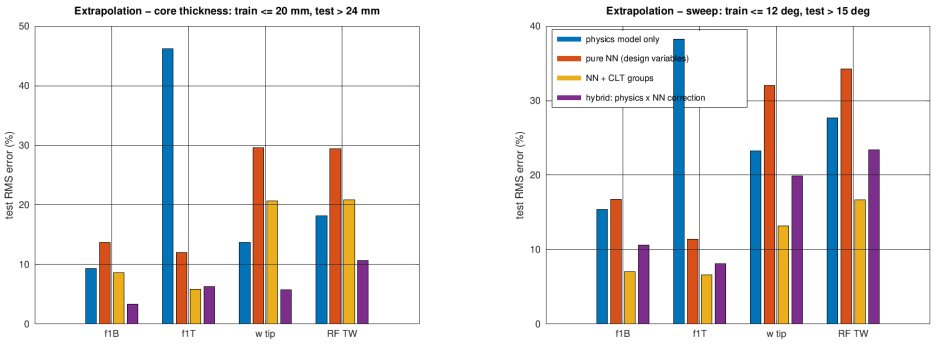
Interpretation:

- **Data efficiency.** The hybrid with **40** FE runs is as accurate as, or better than, the pure network with **240** runs, which is **6× fewer** expensive simulations. At 240 designs, its tip-deflection error is 5× smaller (4.1 % vs 19.8 %).
- **Physically meaningful inputs matter by themselves.** Replacing raw angles by the dimensionless CLT groups (NN + CLT) halves the pure network's error. The bend-twist coupling terms D_{16} and D_{26} are exactly what the beam model misses.
- **Small-data safety.** With 20 designs, the pure network is useless (64 % tip-deflection error). The hybrid is never much worse than the physics model, because its correction starts at zero.

6.3 Extrapolation — designs outside the training range

Test	Physics only	Pure NN	NN + CLT	Hybrid
Core: train ≤ 20 mm, test > 24 mm	9.3 / 46.3 / 13.7 / 18.1	13.7 / 12.0 / 29.6 / 29.4	8.6 / **5.9** / 20.6 / 20.8	**3.3** / 6.2 / **5.8** / **10.7**

Test	Physics only	Pure NN	NN + CLT	Hybrid
Sweep: train ≤ 12°, test > 15°	15.4 / 38.3 / 23.2 / 27.7	16.7 / 11.4 / 32.1 / 34.2	**7.0 / 6.6 / 13.2 / 16.7**	10.6 / 8.1 / 19.9 / 23.4



Extrapolation

This is the most instructive result of the milestone:

- **Core-thickness extrapolation: the hybrid wins clearly.** Error is 3–11 %, versus 12–30 % for the pure network. Core thickness is a variable the *physics model handles correctly* (sandwich bending ∝ h^2, Dunkerley shear). The physics carries the prediction outside the data, and the network only corrects a smooth residual.
- **Sweep extrapolation: the hybrid loses to NN + CLT.** Sweep is precisely what the beam model **ignores**. Its error grows with sweep (15–38 %), so the hybrid must extrapolate a *growing* model-form error, and a network cannot extrapolate a trend it has not seen.

> **Design rule for physics-AI hybrids:** a hybrid extrapolates well in the directions its physics model represents, and no better than its physics model in directions it ignores. The physics core must contain the mechanisms along which you intend to extrapolate. For a swept-wing study, add bending-torsion coupling to `beam_prior`.

This rule will guide Milestone 6: the twin's own physics core is the full FE model, which contains *all* mechanisms.

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7. How Milestone 5 maps to the workflow's Step 6

Step-6 requirement	Delivered
Train ML/DL models (PINN, ANN, GNN)	ANN: PI-Net multi-task network (7 + 1 + 400 + 1 + 1 outputs) and MLP surrogates, all from scratch with verified gradients. PINN principle: discrete governing-equation residual in the loss (§2.5). Graph structure: the 400-element map head works on the FE mesh with a mesh-based evidence prior. A message-passing GNN was not needed: the decoder J already encodes the mesh connectivity *exactly* through K^{-1}.
Embed governing equations / BCs in the loss	L_phys uses eigenvalue perturbation of Kφ = λMφ and adjoint statics of Ku = F. It includes the elastic-

Step-6 requirement	Delivered
	root BC (J_root) and the sensor operator.
Enforce physical consistency	Physics residual reduced by 13–28 %; consistent class/face/map/extent; the hybrid falls back to physics; physical output links (softplus extent ≥ 0, softmax map).
Validate the models	Leakage-free split by wing; test set used once; finite-difference gradient checks (V12); decoder vs nonlinear twins; ablations with 2–3 seeds.

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8. Summary

Key results:

- **Detection:** PI-Net raises the detection rate at an operational false-alarm level over the best Milestone 4 detector: fibre breaks 50 → **73 %**, matrix cracks 18 → **27 %**, debonds 81 → **86 %**.
- **Diagnosis:** 7-class accuracy 0.355 (chance 0.143). Fibre breaks and debonds are recognised; the four low-signal states form a physically explainable confusion block.
- **Localisation:** fibre breaks to 8 mm (full-field) and 37 mm (60 FBG). Core crush, which the rule-based method cannot localise, improves from 161 to 99 mm.
- **Hybrid surrogate:** 2–8 % error with 240 FE runs. It matches a pure network that needs 6× more data, and it extrapolates in core thickness with 3–11 % error versus 12–30 %.
- **Most useful physics ingredients (in order):**
    1. healthy-referenced inputs (+40 % accuracy);
    2. reduced-order mechanics with discrepancy learning;
    3. dimensionless CLT inputs;
    4. the evidence prior for localisation;
    5. the physics loss, which buys consistency and a small low-label benefit.

Honest limits:

- The static-strain damage signal (0.4–2.6 με) is near the sensor noise. This bounds every model.
- Delamination is undetectable by this sensor network.
- Direct regression of strength from features is unreliable, so Milestone 6 computes strength by **FE model updating** instead.
- The hybrid surrogate inherits the blind spots of its physics core (sweep).
- Results vary by about ±0.01 between runs because of multithreaded floating-point summation, in Octave and MATLAB alike. The conclusions do not change.

Next — Milestone 6 (Steps 7–10):

- Assemble the digital twin: FE model, PI-Net diagnosis, Bayesian model updating of the damage, and virtual sensing (strain and RF at unmeasured locations).
- Damage quantification by model updating, then fatigue-driven prognostics and RUL.
- End-to-end validation on unseen damage scenarios.

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